

April 2026



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**COLLEGE OF SCIENCE
AND ENGINEERING**

3 Questions. One Answer.

01

SURGE

What

it is

A VLAN-segmented lab network that gives SURGE a realistic, enterprise-grade environment to operate in.

How

it works

An automated agent pipeline scans the network, identifies systems, correlates vulnerabilities, and scores risk in real time through a live dashboard.

Why

it matters

Manual assessments take hours. SURGE automates the process, making security analysis faster, more accessible, and just as effective.

What is SURGE?

Multi-Agent System

SURGE coordinates distributed AI agents that analyze network activity, detect threats, and identify vulnerabilities across segmented environments.

Network-Wide Analysis

Agents work across VLAN 10 (Attackers) and VLAN 20 (Users) to discover hosts, services, and risks throughout the network.

Data Analysis & Insights

Network data—topology, ports, OS fingerprints, and CVEs—is analyzed to surface risks and patterns in a clear, usable way.

How Does SURGE Work?

03

DEPLOY

Network Team

Built a VLAN-segmented lab network with Cisco switches and an ISR4311 router (ROAS), and 5 Raspberry Pis simulating end devices.

Configure DHCP and NAT/PAT

Then deploy SURGE into a controlled test environment.

ANALYZE

AI Agents

Recon agent scans across subnets, identifies systems and vulnerabilities, and uses ML to score and prioritize risk.

VISUALIZE

Dashboard

Render a live network map as scans progress, color-code nodes by severity, and stream agent activity in real time.s in real time.

SECURE

Report Output

Reporter agent generates tailored reports for executive, technical, and public audiences, each with actionable remediation steps.

Building the Network

04

Cisco 2960-24TT Switches ×2

Layer 2 Cisco 2960 switches segment the network and carry VLAN traffic using an 802.1Q trunk link

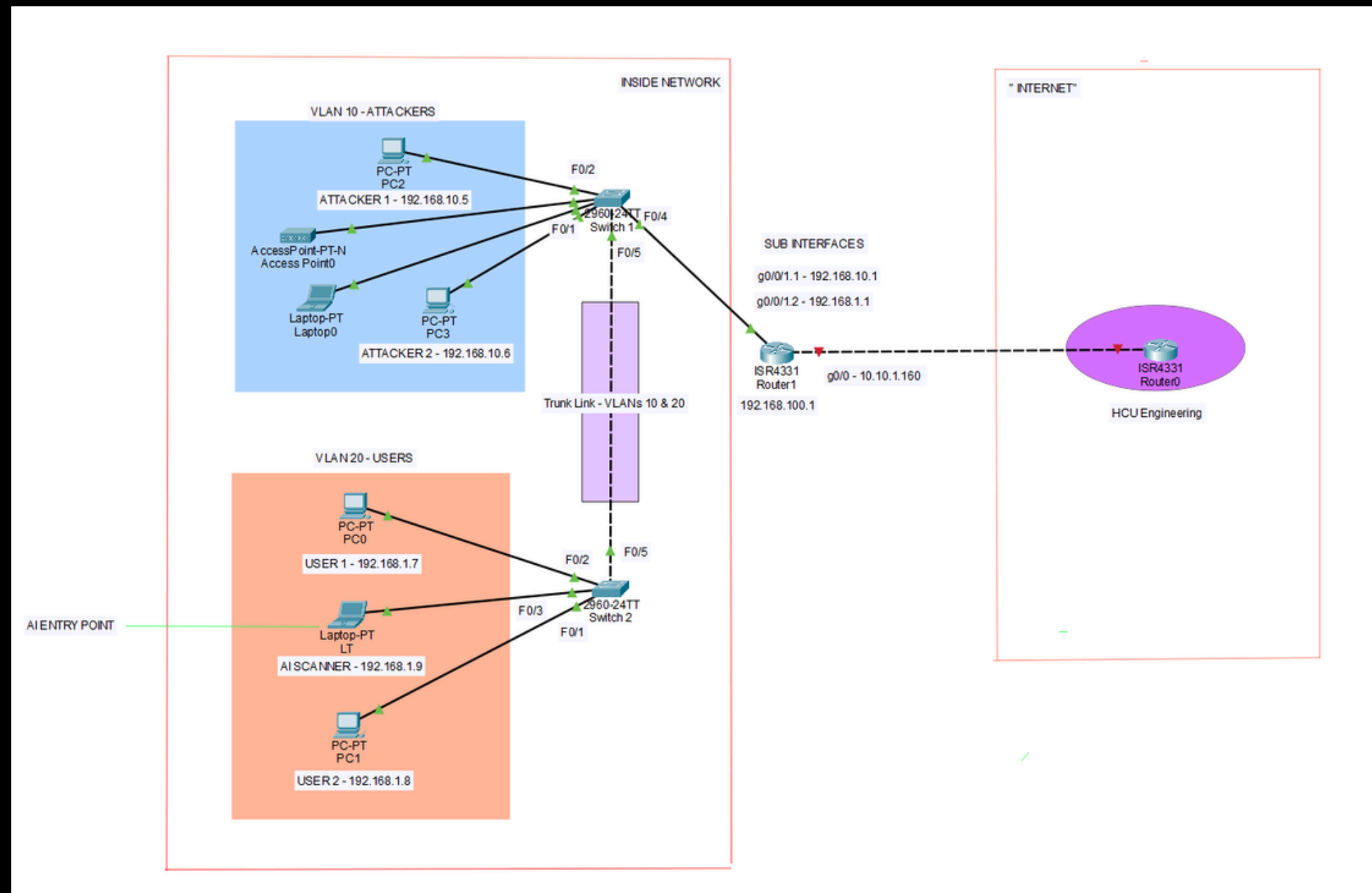
Cisco ISR4331 Router ×2

Cisco ISR4331 routes traffic between VLANs using router-on-a-stick and handles NAT for external access.

WaveLink Wireless AP

Wireless access point extending lab network connectivity. Provides Wi-Fi access for devices within the VLAN environment.

Lab Network Topology



Why SURGE Matters

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1

Networks are becoming more complex

Modern enterprise environments span VLANs, subnets, firewalls, and NAT — a realistic test network must reflect that.

2

Manual analysis is slow and inefficient

Analysts waste hours on network discovery and scan configuration before any real vulnerability work begins.

3

Threats are harder to detect

Attackers hide across network boundaries. A flat lab network wouldn't expose Surge to these real-world conditions.

4

Security insights are not always easy to understand

Reports must be meaningful — and that requires a structured, realistic target environment to produce accurate findings.

Our Answer

- Simulate real enterprise network complexity
- Eliminates manual analysis with automation
- Surfaces threats across network boundaries
- Turns findings into clear, actionable insights

Network Add-ons

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→ NAT Overload (PAT)

All internal 192.168.0.0/16 traffic translates through external interface 10.10.1.160 via Port Address Translation. Validated: 65 active translations, 3,986 hits, 0 misses.

◇ DHCP Pools

Router provides dynamic IP assignment for both VLANs. Pool excludes gateway address. DNS resolves through 8.8.8.8. Mirrors real-world enterprise DHCP deployment.

⬡ Router-on-a-Stick

Single physical router port split into subinterfaces: g0/0/1.1 for VLAN 10 and g0/0/1.2 for VLAN 20. Enables controlled inter-VLAN routing without Layer 2 broadcast floods.

● Tailscale VPN

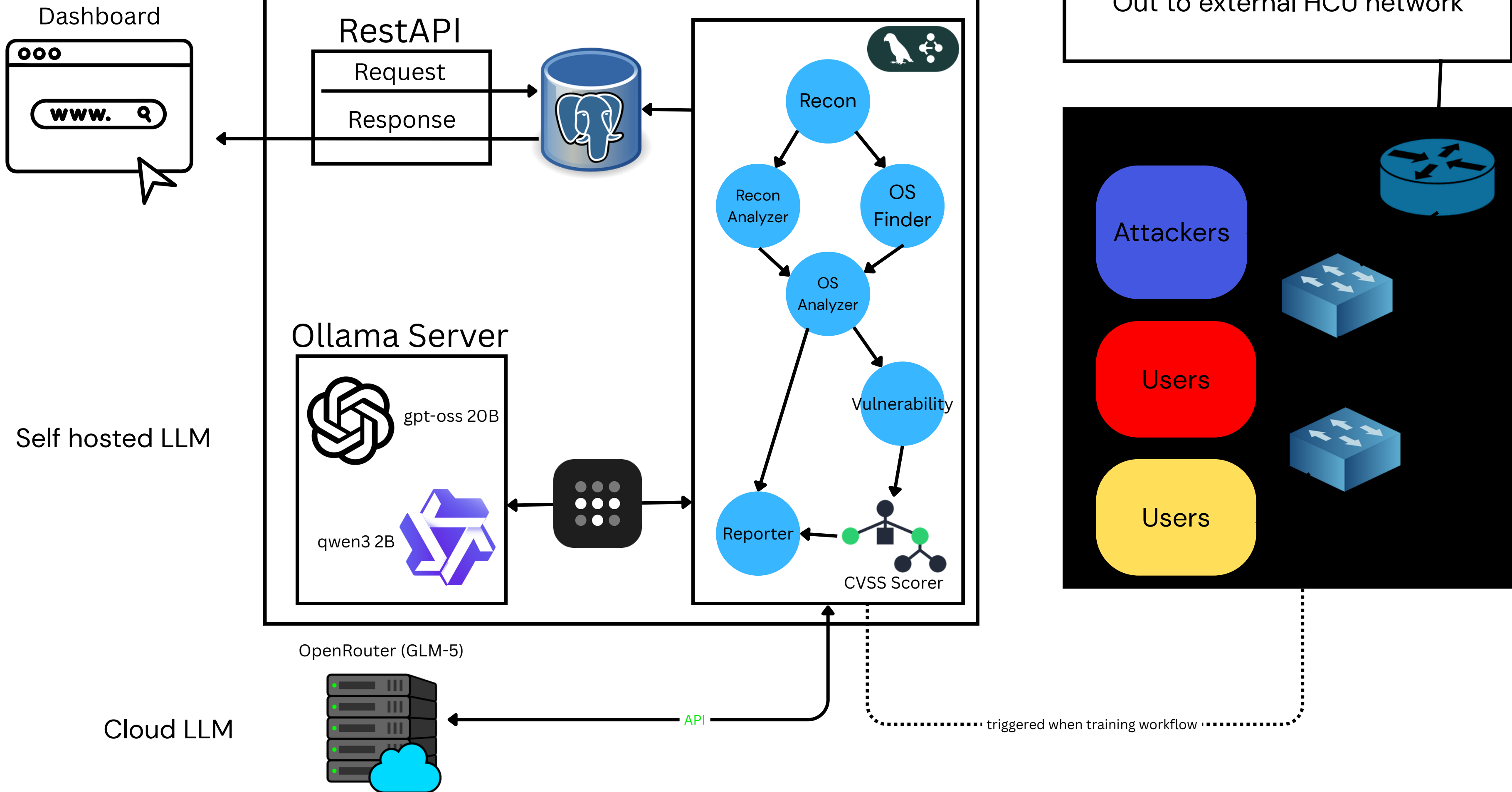
Secure remote access to the GPU server from off-campus. Enabled the dev team to test Surge against the lab network remotely throughout development.

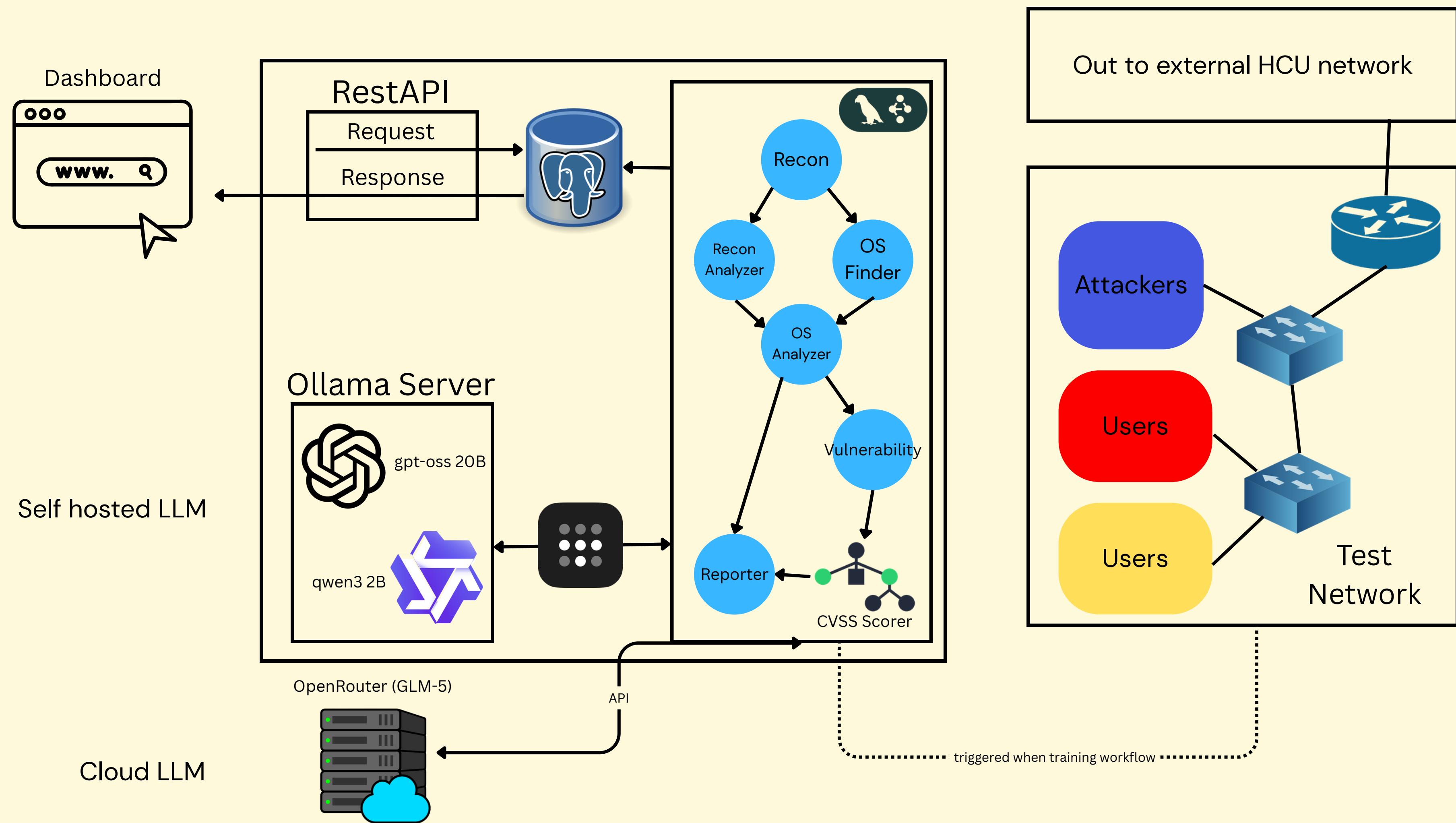
▣ Future: ACLs

Access Control Lists between VLANs are the planned next phase. Will restrict which services each VLAN can reach, adding a Layer 3 enforcement boundary above the current ROAS config.

⊛ Future: IDS/IPS

A monitoring device on the trunk link would inspect all inter-VLAN traffic in real time. Planned placement between Switch 1 and the router for full east-west visibility.





Introducing our Agents

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Agents are software that act on their own to complete a specific task

Surge grew from 1 agent to 7 agents

Each agent plays its part in analyzing the network



Each Agent Falls Into One of the 3 Categories

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Scanners



Reporters



AI Modeling / Data Analytics



The Scanner Agents

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Recon Agent: Scans the main entry point to the network, attempting to map all devices and endpoints by repeatedly scanning the network.



OS Finder Agent: Scans every host and open port found from the Recon agent to identify their OS and services.

The AI / Data Agents

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Vulnerability Finder Agent: Parses the OS Finder results and uses the CIRCL API to look up CPEs and banners to identify what hosts and OSs on the network have active vulnerability reports.

CVSS Scorer Agent: Uses a regression model to give a vulnerability score to all hosts identified as vulnerable.



The Reporter Agents

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Recon Analyzer Agent: Summarizes all network findings, such as what ports are open, gives key observations, and suggests possible reconnaissance next steps.

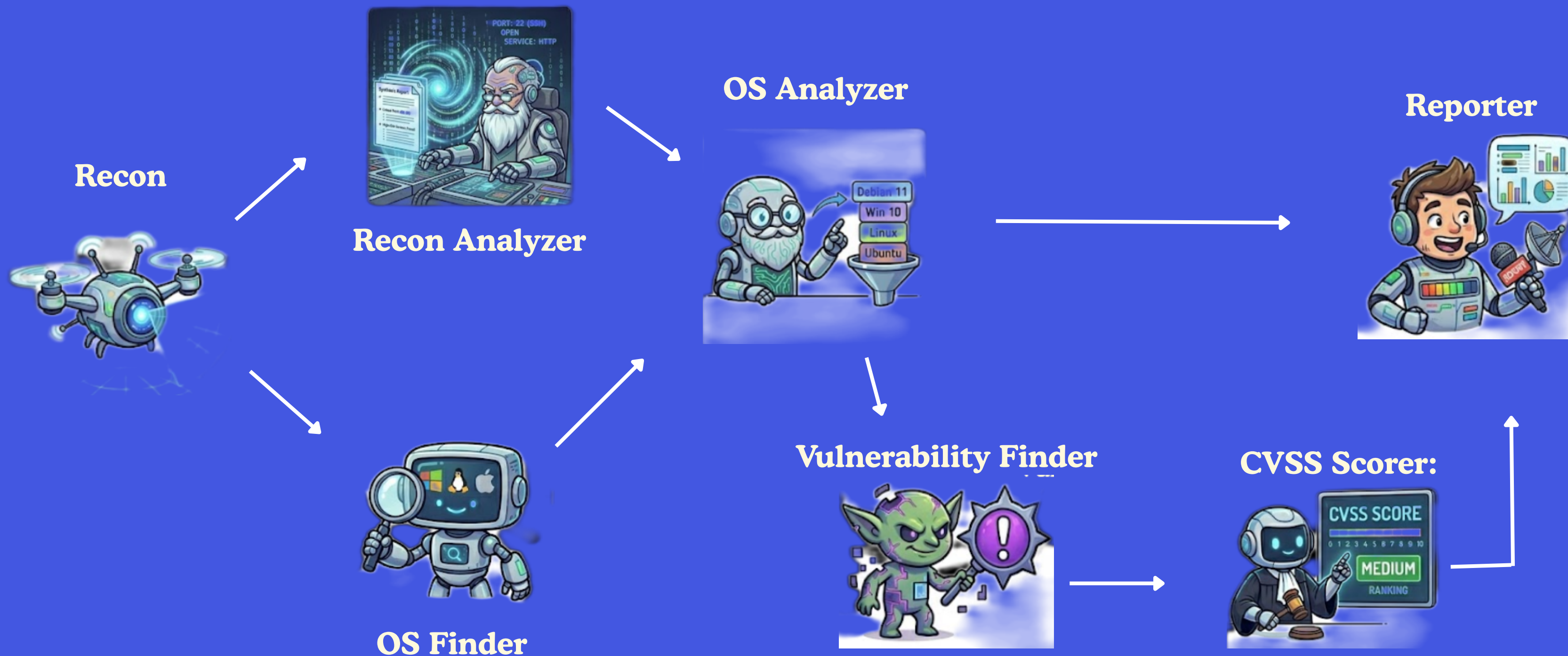
OS Analyzer Agent: Builds a tabular report of OSs, vendors, hosts, and CPE highlights from Recon's and OS Finder's network scans.



Reporter Agent: Takes all the knowledge from all agents and creates a non-technical report, a technical report, and a media report.

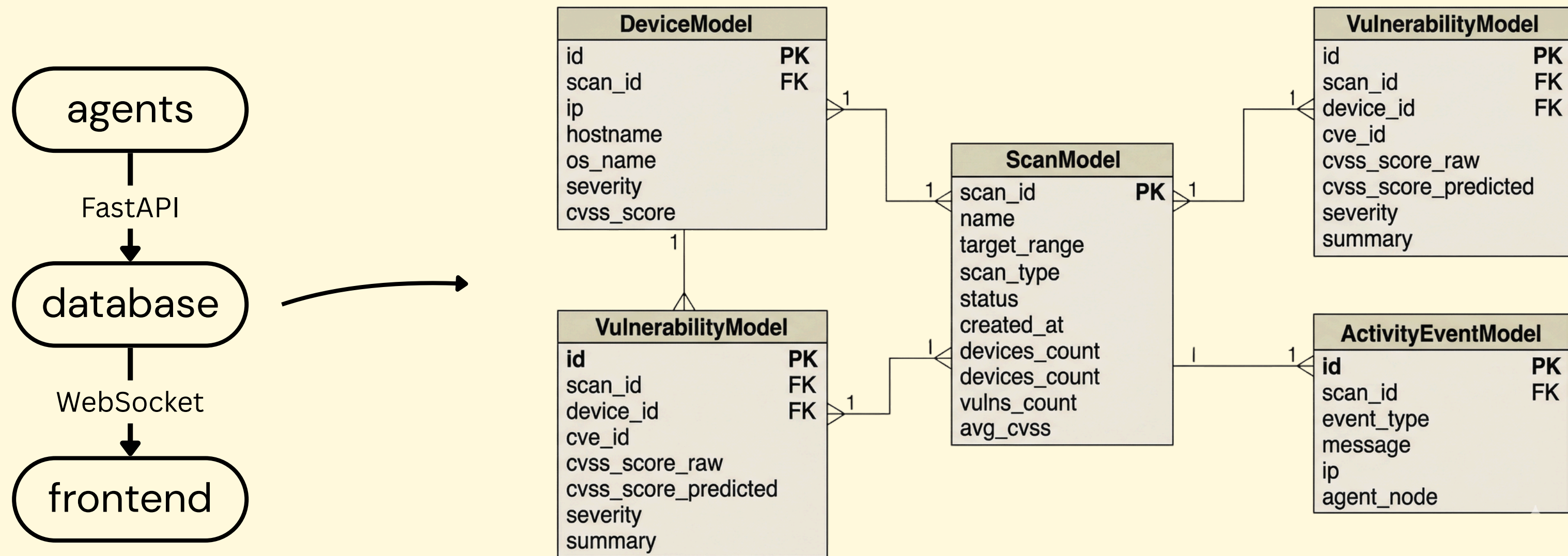
Full Workflow

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Data Collection: How We Store It

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Why We Self-Hosted

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Goal:

Run high-performance LLMs locally for faster, secure data processing

Original Plan:

- Use a **vLLM** (*high-performance inference engine*)
- Limitation: Requires newer GPUs (**Ampere, compute 8.0+**)

Our Approach:

- Switched to **Ollama + GGUF** quantization (**Q4_K_M**)
- Compatible with a wider range of **CUDA GPUs**

Result:

- Efficient local inference without hardware constraints
- Models fit within available VRAM:
 - **Qwen3-32B (~20GB)**
 - **GPT-OSS-20B (~14GB)**

Testing With the Cloud First

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Why Cloud Testing:

- Local GPU was set up, but we wanted faster iteration during development

What We Used:

- OpenRouter for instant access to cloud models (GLM-5, DeepSeek)

How It Worked:

- One flag: `USE_ONLINE=1` swaps the entire inference layer
- Seamless switch between local and cloud environments

Development Setup:

- Tested remotely over Tailscale from off-campus

Result:

Fast experimentation
with minimal cost

Total spent: **\$3.43**

Two GPUs, Two Models, One Pipeline

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Architecture:

- GPU 0 → Qwen3-32BVulnerability analysis, OS analysis, reporting
- GPU 1 → GPT-OSS-20B Recon decisions, data formatting

Parallel Execution:

- GPUs are isolated → no PCIe contention
- Both models run simultaneously for efficiency

Scalability:

OpenRouter handles overflow when both GPUs are busy

Optimization:

- Context windows reduced from 128K → 8K for structured tasks
- Maintained quality while significantly improving speed

Our Dashboard in 6 Parts

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	DASHBOARD	SCANS	TOPOLOGY	EXPLOITS	REPORTS	AGENTS
What it shows	Network overview, device list, activity feed, vuln chart	All past and active scans with status and duration	Full-screen interactive network graph	Complete CVE list with severity and exploit status	Generated security reports by audience	Live agent pipeline status and configuration
Key feature	Auto-switches between Live, Latest, and History mode	WebSocket progress bar per agent node	D3 force-directed graph with pan and zoom	Filterable by severity, exploit availability, and status	4 report types: Executive, Technical, Public, Final	Shows which agent is running and % complete

Q/A

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Thank You